HelpHub - Design Documentation

Team Members: Abhinay Sharthi, Emily Huang

HelpHub is a multi-tenant service desk where multiple organizations use the same application but see only their own data. A company signs up as a tenant and invites its agents and customers with share codes. Customers submit tickets on the website, or other systems send them via an API key; an AI sorts each ticket by the company's rules and automatically assigns it to an agent. Agents work a queue, reply, add internal notes, escalate tickets to the admins, and complete them, while admins track the team on a dashboard.

User Personas:

Rhea, a Support Agent: Rhea works in support at a 60-person logistics company. Issues arrive by email to a shared inbox, through the website contact form, and occasionally from internal tools via API. She switches between three places and still loses track. She wants one queue where every request has an owner and a status, no matter how it arrives.

Daniel, a Tenant Admin: Daniel runs the support team and spends a lot of time organizing the support tickets. He wants the internal order system to automatically open a ticket when a shipment fails, without having to open one manually.

Sabrina, a QA Lead: Sabrina wants numerical data on which aspects of their services have the most bugs to do better-informed testing for future features. She wants to analyze bug distribution by priority and category to inform better testing strategies.

Priya, a Customer: Priya has a problem with her account on a website. Priya wants to resolve this problem quickly with as little hassle as possible, over the phone or online.

User Stories:

Abhinay Sharthi:

1. As Daniel, I want to register my organization as a tenant and invite agents via a share code so that my team has an isolated workspace.
2. As Priya, I want to register and log in securely so that my account and data are private.
3. As Sabrina, I want to view a performance dashboard (ticket counts by status, response time, resolution time) and see analytics by priority, category, and quarter, with a CSV export, so that I can track team performance and identify improvement areas.
4. As Daniel, I want to generate API keys so that internal systems (order management, shipment tracking) can create tickets programmatically.

Emily Huang:

1. As Rhea, I want one queue of all my tickets, whether they came from the website or another system through an API key, sorted by priority or date, so that I don't switch between tools.
2. As Rhea, I want to view ticket details, change status, assign agents, and post public/internal comments so that I can manage the entire ticket lifecycle.
3. As Rhea, I want to escalate a ticket with a reason and notes so that the ticket transfers with full conversation history to the next support tier.
4. As Priya, I want to submit a support ticket via the web portal and view my tickets' status in real-time so that I can track resolution without calling.
5. As Priya, I want to reply to agent comments on my ticket and close it when it's resolved, so that I can follow up without re-explaining my issue.

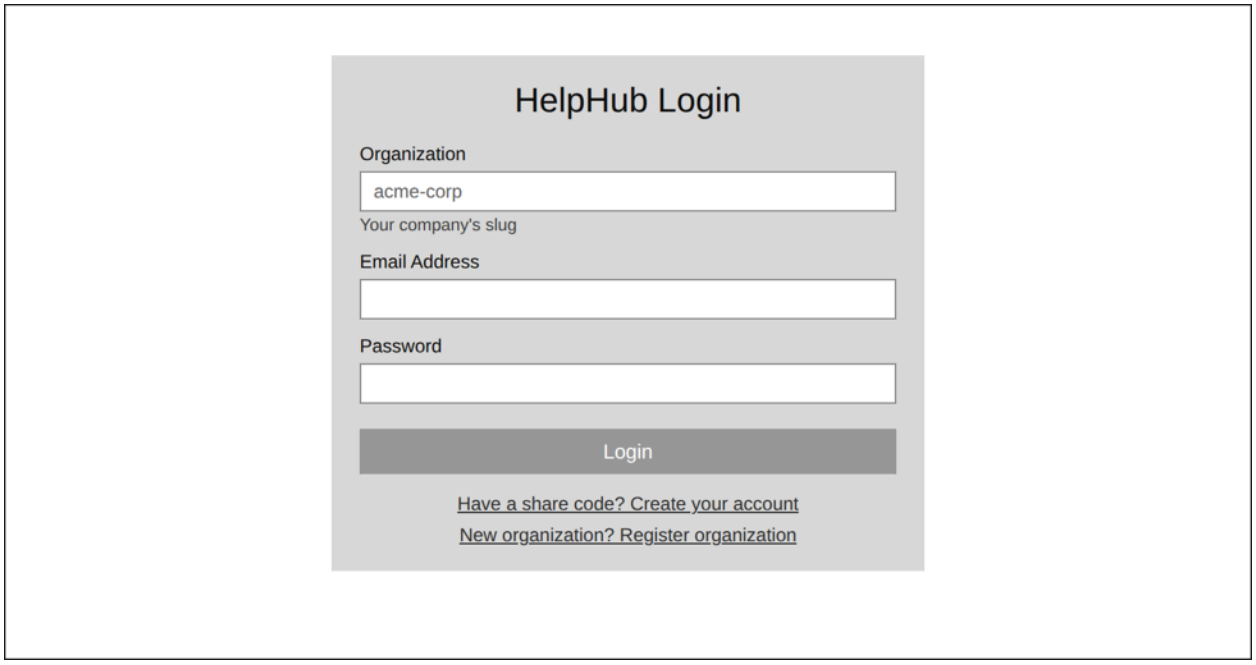
Work Division:

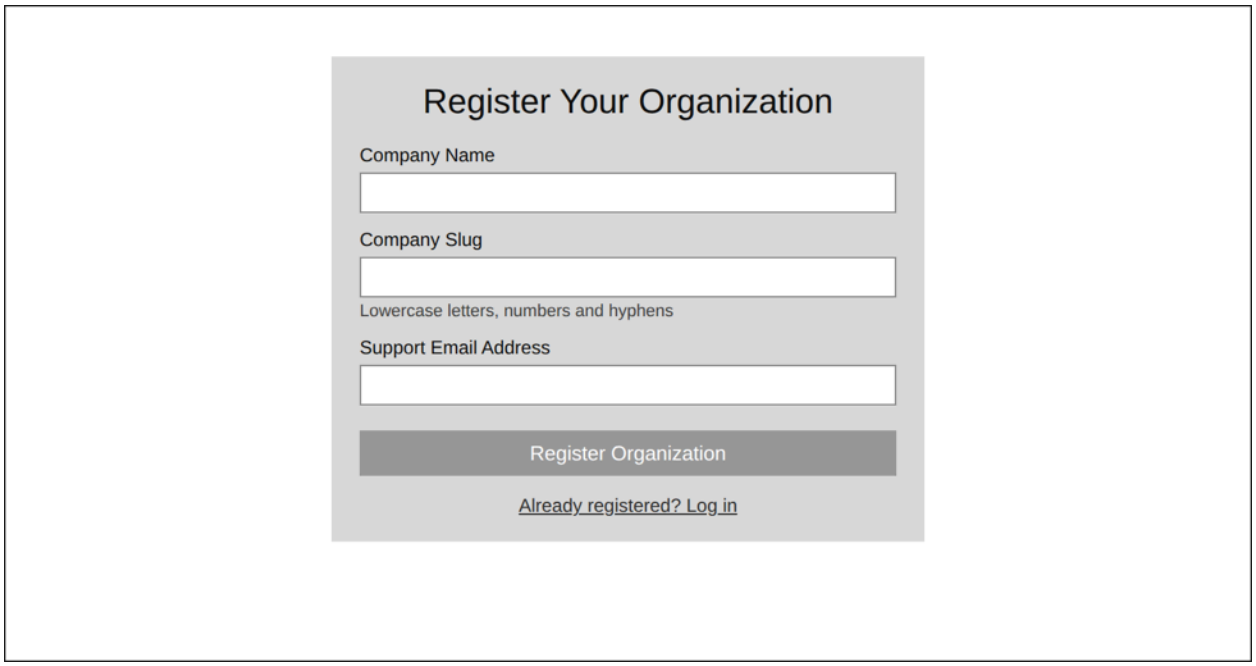
Abhinay Sharthi — tenants, users, API key collections. Tenant registration, user login/logout, agent management dashboard, SLA configuration, admin performance dashboard, analytics dashboard, API key generation and revocation, tenant resolution middleware

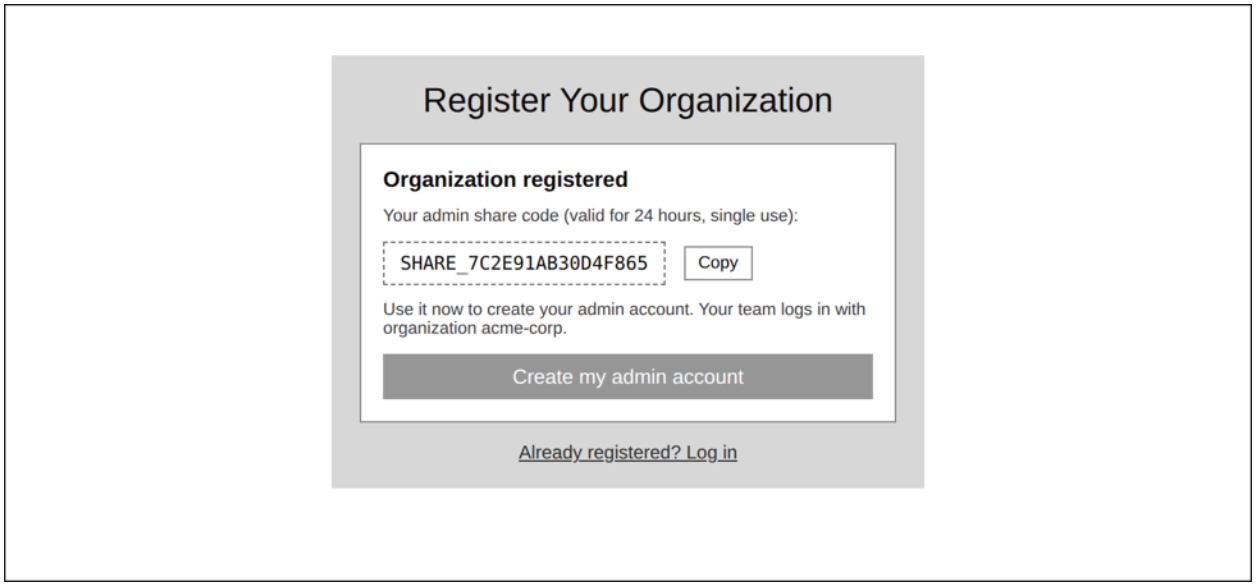
Emily Huang — tickets, comments, auditLog collections. Ticket CRUD, queue with filtering, ticket detail view, public/internal comments, escalation with context transfer, SLA computation, customer portal, audit log creation

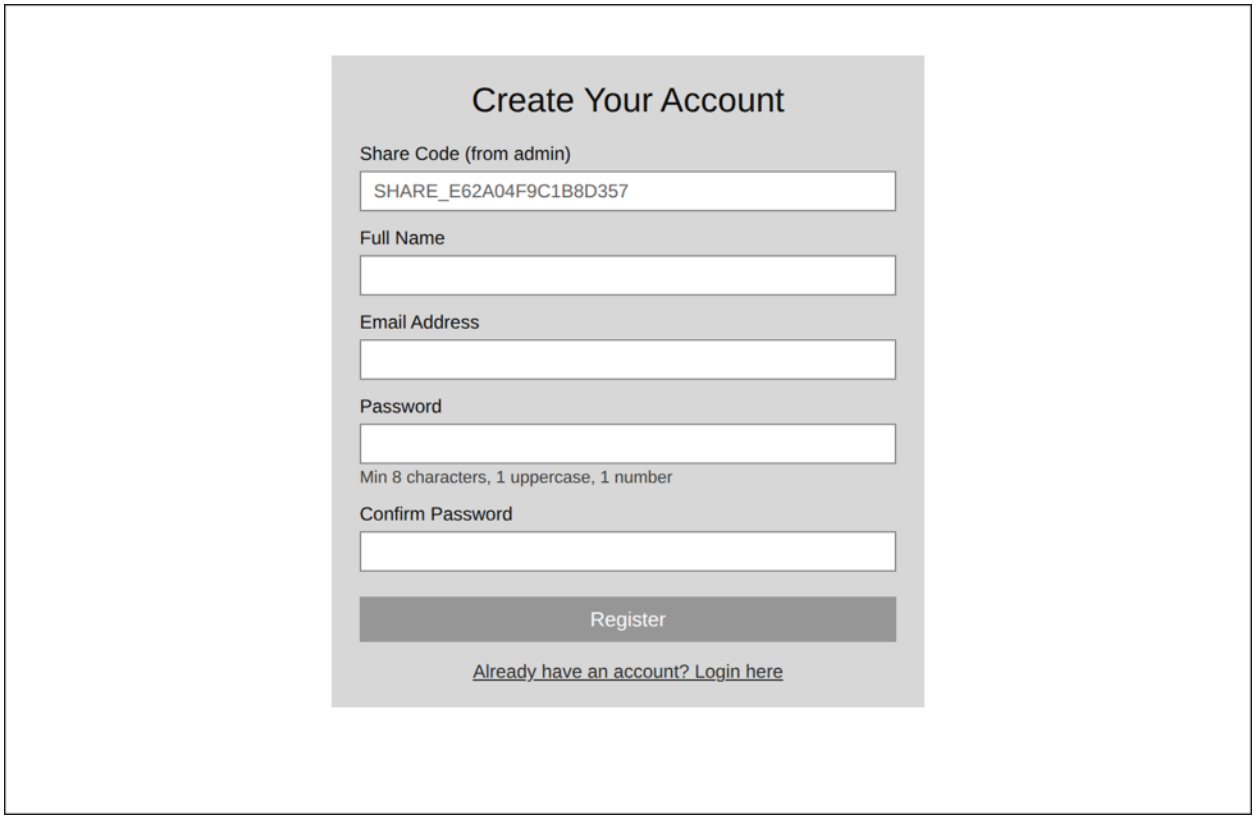
Design MockUps:

Some design changes during building.
Sign-up and login:

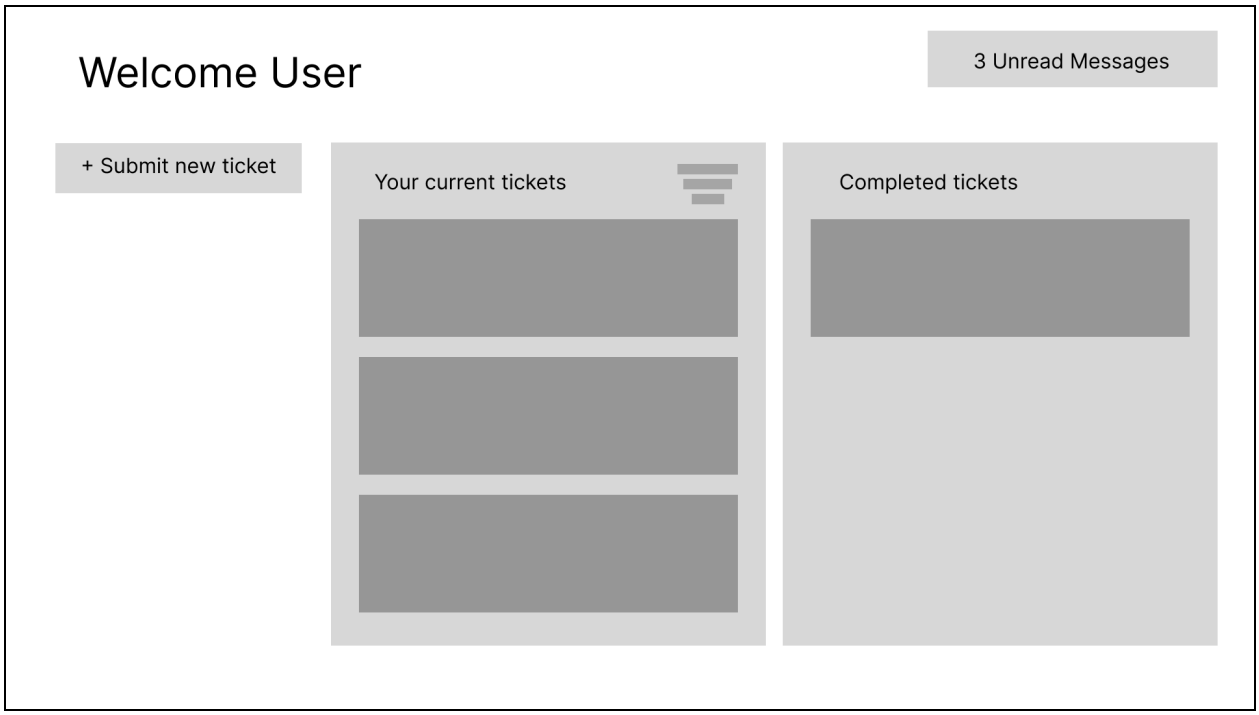

HelpHub Login
Organization
acme-corp
Your company's slug
Email Address
Password
Login
Have a share code? Create your account
New organization? Register organization


Register Your Organization
Company Name
Company Slug
Lowercase letters, numbers and hyphens
Support Email Address
Register Organization
Already registered? Log in

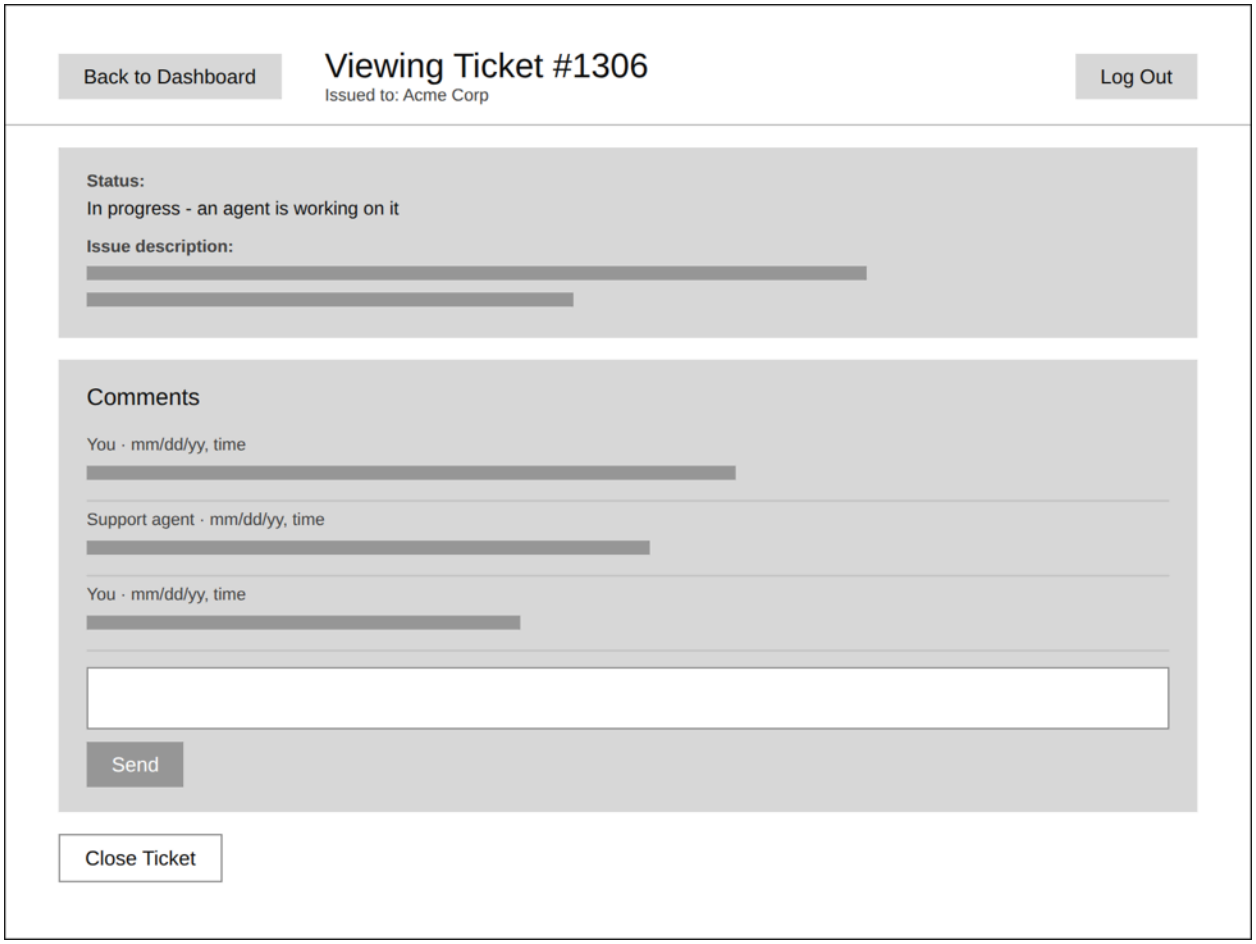




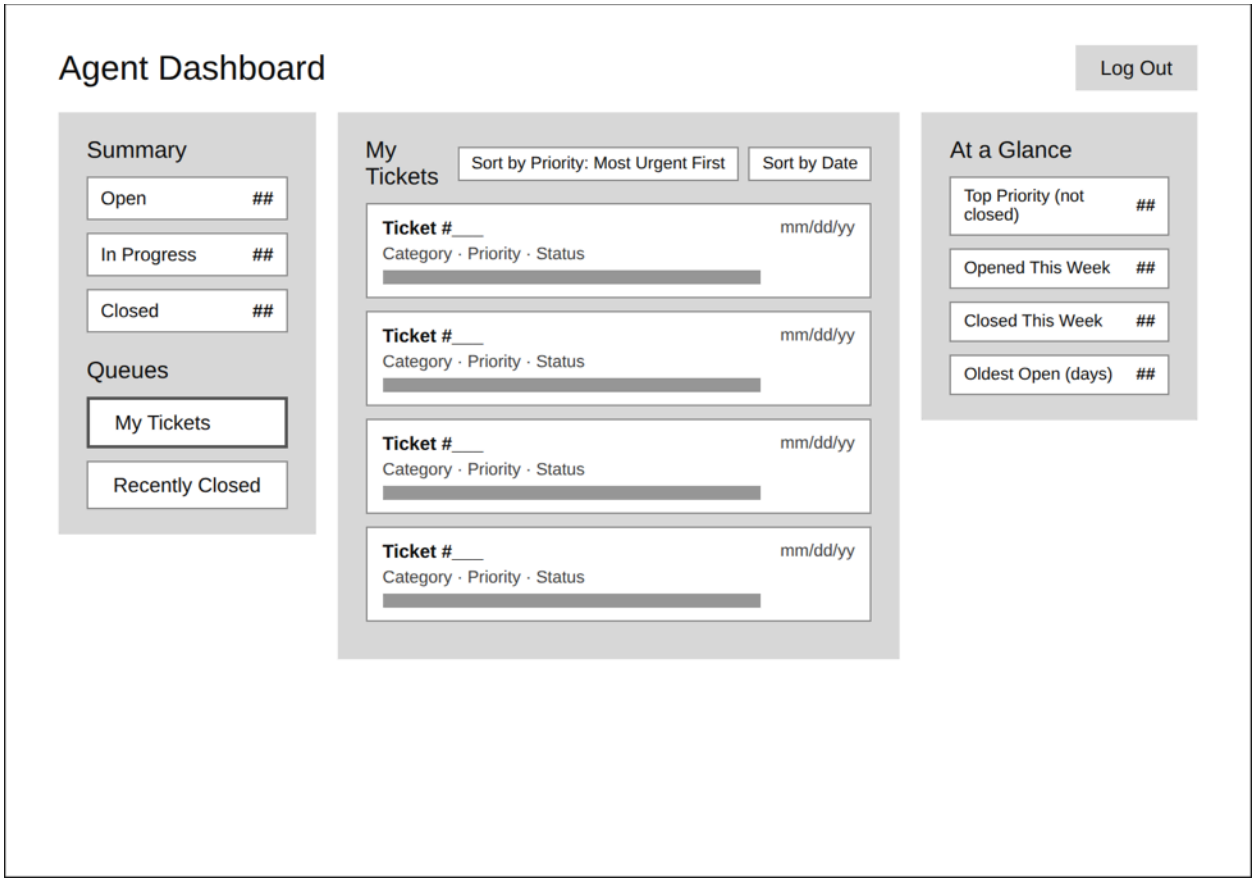
Customer Portal:

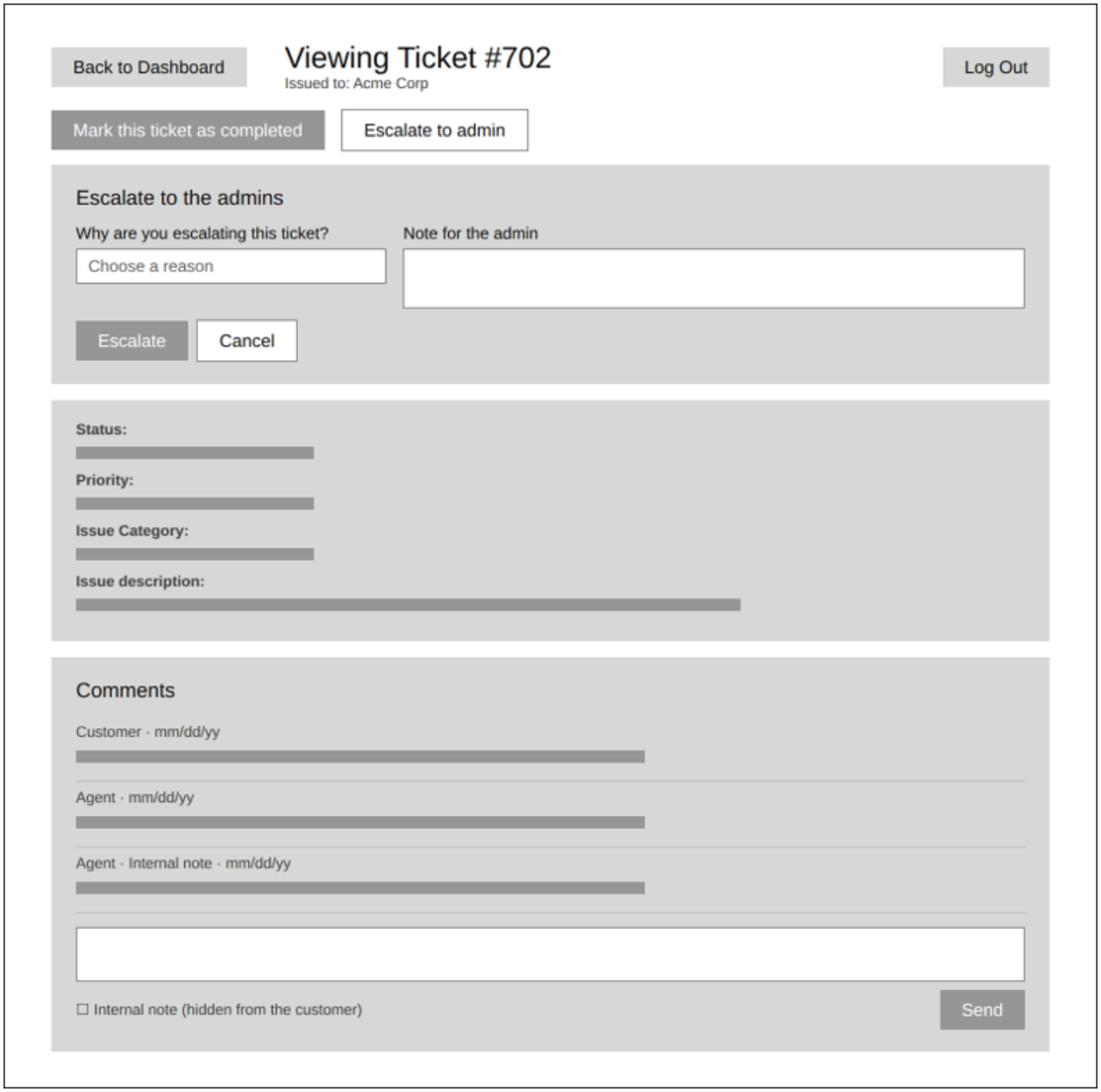

Welcome User
3 Unread Messages
+ Submit new ticket
Your current tickets
Completed tickets


Help Ticket Submission Form
Type of Issue
Description
Type here...
Submit

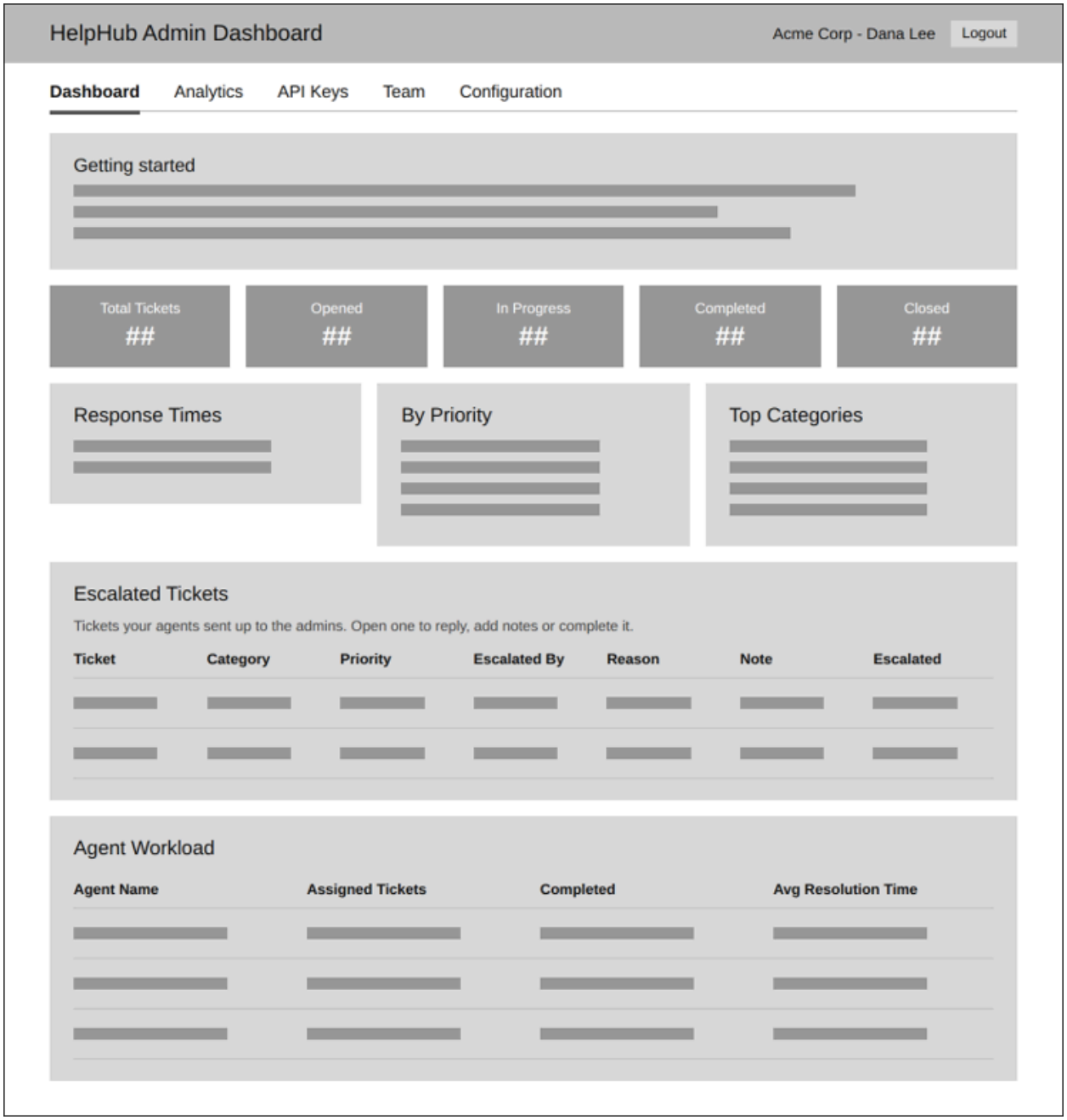

Back to Dashboard
Viewing Ticket #1306
Issued to: Acme Corp
Log Out
Status:
In progress - an agent is working on it
Issue description:
Comments
You · mm/dd/yy, time
Support agent · mm/dd/yy, time
You · mm/dd/yy, time
Send
Close Ticket

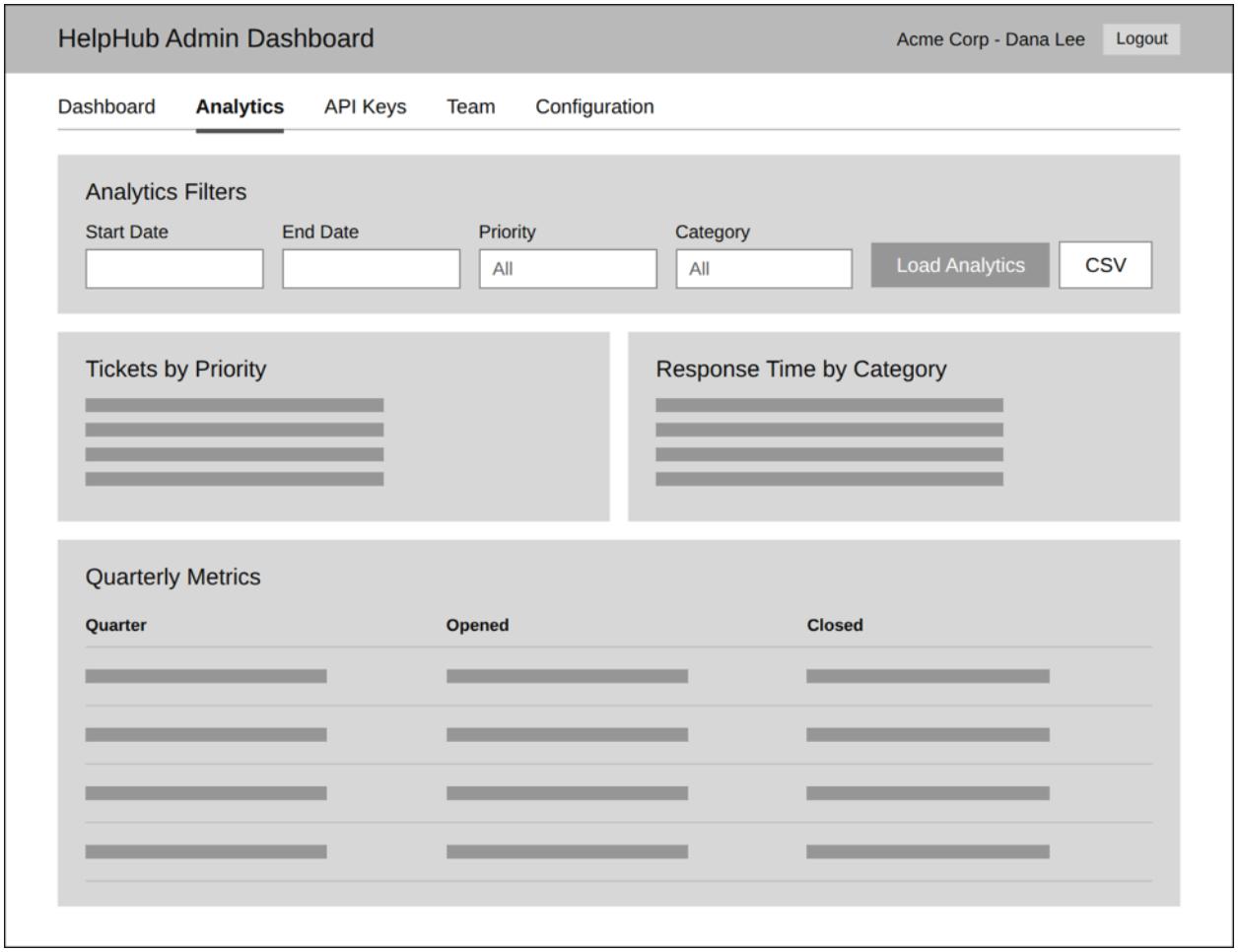
Agent Pages:

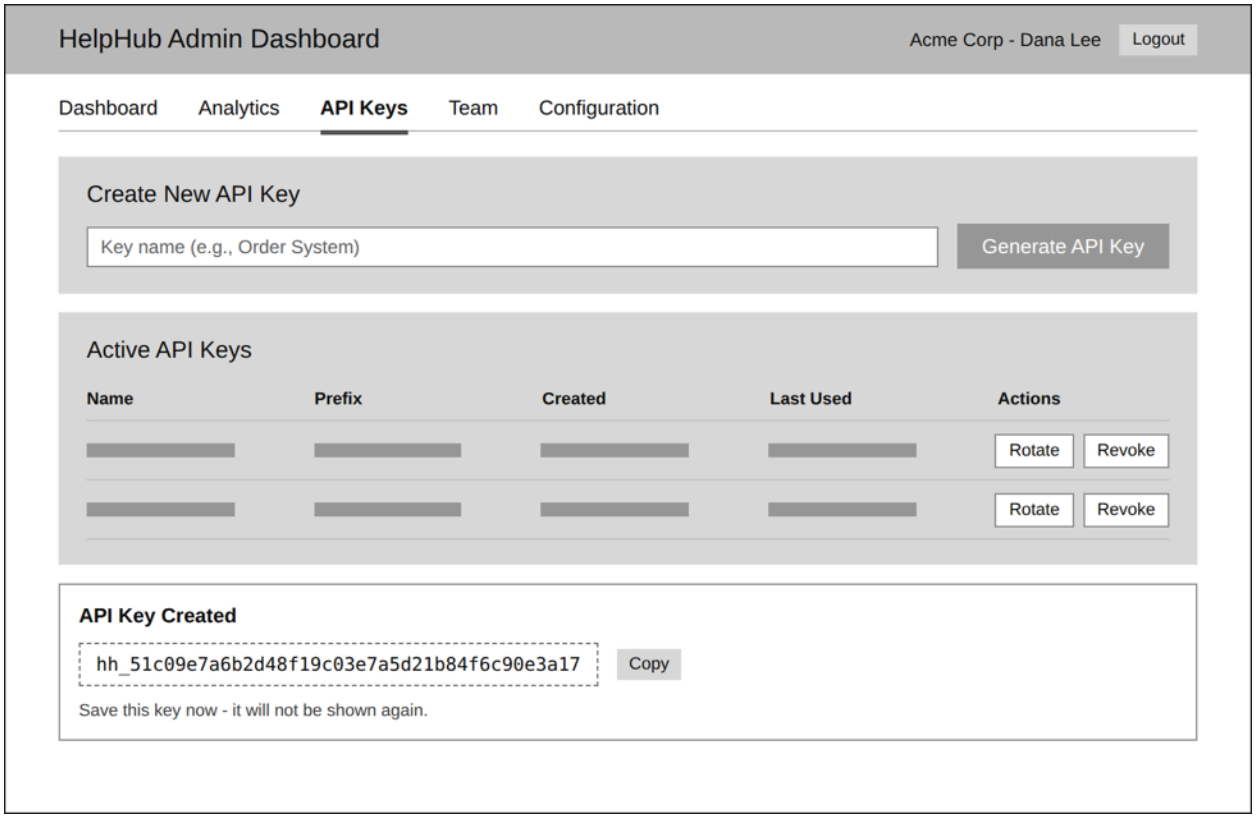

Agent Dashboard
Log Out
Summary
Open ##
In Progress ##
Closed ##
Queues
My Tickets
Recently Closed
My Tickets
Sort by Priority: Most Urgent First
Sort by Date
Ticket #___
mm/dd/yy
Category · Priority · Status
Ticket #___
mm/dd/yy
Category · Priority · Status
Ticket #___
mm/dd/yy
Category · Priority · Status
Ticket #___
mm/dd/yy
Category · Priority · Status
At a Glance
Top Priority (not closed) ##
Opened This Week ##
Closed This Week ##
Oldest Open (days) ##

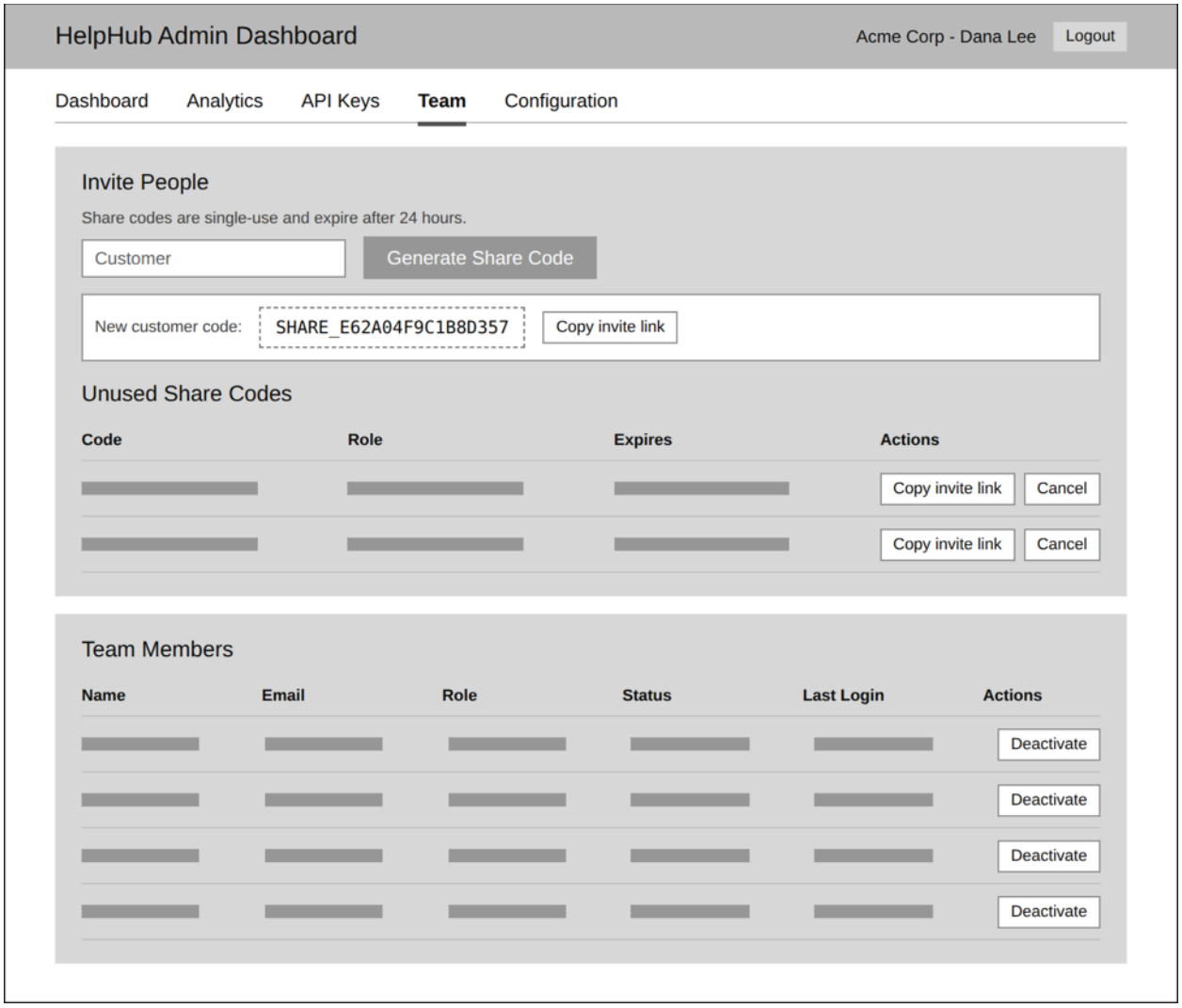


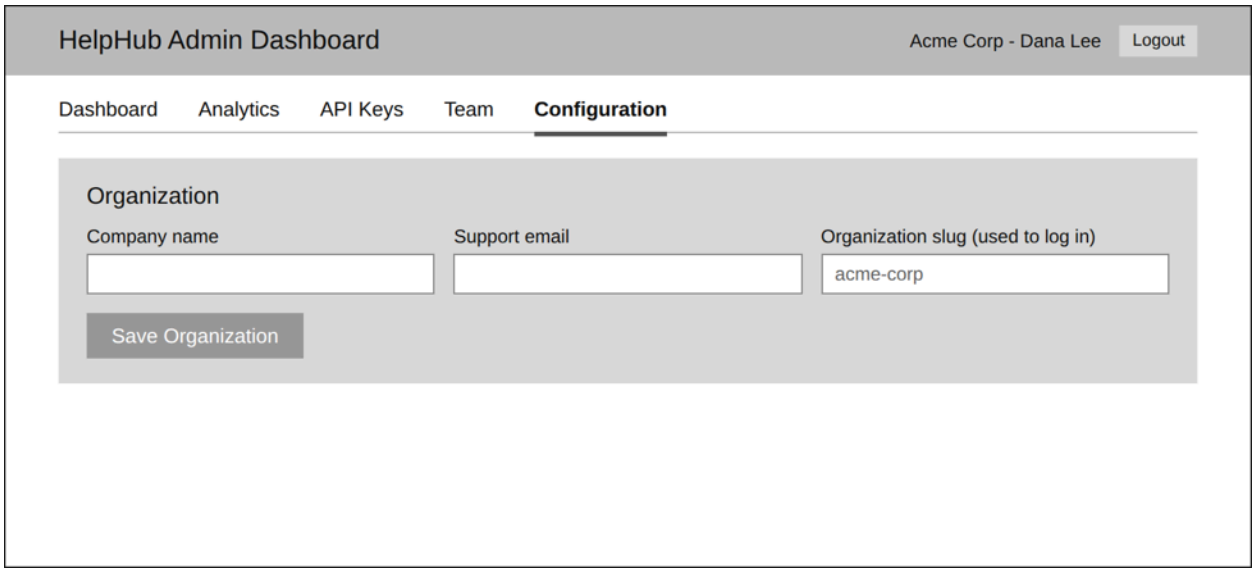
Admin dashboard:


HelpHub Admin Dashboard
Acme Corp - Dana Lee
Logout
Dashboard
Analytics
API Keys
Team
Configuration
Getting started
Total Tickets
##
Opened
##
In Progress
##
Completed
##
Closed
##
Response Times
By Priority
Top Categories
Escalated Tickets
Tickets your agents sent up to the admins. Open one to reply, add notes or complete it.
Ticket
Category
Priority
Escalated By
Reason
Note
Escalated
Agent Workload
Agent Name
Assigned Tickets
Completed
Avg Resolution Time


HelpHub Admin Dashboard
Acme Corp - Dana Lee
Logout
Dashboard
Analytics
API Keys
Team
Configuration
Analytics Filters
Start Date
End Date
Priority
All
Category
All
Load Analytics
CSV
Tickets by Priority
Response Time by Category
Quarterly Metrics
Quarter
Opened
Closed


HelpHub Admin Dashboard
Acme Corp - Dana Lee
Logout
Dashboard
Analytics
API Keys
Team
Configuration
Create New API Key
Key name (e.g., Order System)
Generate API Key
Active API Keys
Name
Prefix
Created
Last Used
Actions
Rotate
Revoke
Rotate
Revoke
API Key Created
hh_51c09e7a6b2d48f19c03e7a5d21b84f6c90e3a17
Copy
Save this key now - it will not be shown again.


HelpHub Admin Dashboard
Acme Corp - Dana Lee
Logout
Dashboard
Analytics
API Keys
Team
Configuration
Invite People
Share codes are single-use and expire after 24 hours.
Customer
Generate Share Code
New customer code:
SHARE_E62A04F9C1B8D357
Copy invite link
Unused Share Codes
Code
Role
Expires
Actions
Copy invite link
Cancel
Copy invite link
Cancel
Team Members
Name
Email
Role
Status
Last Login
Actions
Deactivate
Deactivate
Deactivate
Deactivate


HelpHub Admin Dashboard
Acme Corp - Dana Lee
Logout
Dashboard
Analytics
API Keys
Team
Configuration
Organization
Company name
Support email
Organization slug (used to log in)
acme-corp
Save Organization